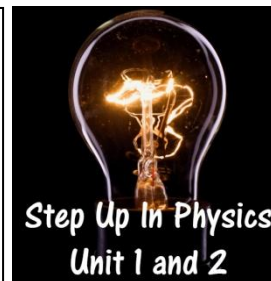


S.H.E: Heating Processes

Problems Worksheet



1. The series of pipes containing a refrigerant in a heat pump is a closed system. It passes through the following stages:
 - Compressor which pressurises the gaseous refrigerant, increasing its temperature
 - Condenser where the pressurised refrigerant condenses
 - Expansion valve where liquid refrigerant has pressure reduced, decreasing its temperature
 - Evaporator where the liquid refrigerant evaporates
- a. Which stage(s) does the internal energy of the refrigerant increase? For each stage you list, describe where the energy comes from.
- b. Which stage(s) does the internal energy of the refrigerant decrease? For each stage you list, describe where the energy goes.
- c. Both the condenser and evaporator are referred to as heat exchangers. A reverse cycle air conditioner is an example of a heat pump. Explain where each heat exchanger should be placed (inside or outside) during summer when the air conditioner needs to cool the house.

4. Describe two features of a car radiator system that make it efficient at losing heat from the coolant.